

BANARAS HINDU UNIVERSITY

B.Com. (Hons.) — Semester IV Examination, 2023-24

Paper No. BCH-226: Business Statistics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. All questions carry equal marks (14 marks each).***Question 1.**

(i) Prepare chain base index numbers from the following data:

Year	2019	2020	2021	2022	2023
Price (Rs.)	80	120	132	264	396

[7]

(ii) From the chain base index numbers given below, prepare fixed base index numbers:

Year	2018	2019	2020	2021	2022	2023
Chain Base Index No.	92	102	104	98	103	101

[7]

OR

Explain the points to be considered in the construction of Index Numbers.

[14]

Question 2. Calculate index numbers for 2020 and 2023 from the following data relating to cost of living of the working class in a city:

Items	Weight	Price Index of items	
		2020	2023
Food	48	110	130
Clothing	8	120	125
Fuel and Light	7	110	120
Room Rent	13	100	100
Miscellaneous	14	115	135

There is a wage increase of 8% from the year 2020 to 2023. Is it adequate?

[14]

OR

Why Fisher's Index Number is Ideal? Explain with the help of imaginary figures.

[14]

Question 3. Calculate trend values by the method of least squares from the data given below:

Year	2017	2018	2019	2020	2021	2022	2023
Value	75	67	68	65	50	54	41

[14]

OR

Explain moving average method for measurement of trend with the help of imaginary figures.
[14]

Question 4. Calculate seasonal index from the following data using the average method:

Year	Quarter			
	I	II	III	IV
2019	72	68	80	70
2020	76	70	82	74
2021	74	66	84	80
2022	76	74	84	78
2023	78	74	86	82

[14]

OR

What are various methods of measuring seasonal variations in time series? Discuss any one method with the help of imaginary figures. [14]

Question 5.

- (i) A total of 10 samples of 8 each was drawn from a universe. The average range (arR) and average standard deviation ($ar\sigma$) of these samples were 0.0811 and 0.0263 respectively. Estimate the standard deviation of the universe from which these samples were drawn when $d_2 = 2.847$ and $c_2 = 0.9027$. [7]
- (ii) If $arX = 0.8768$, $arR = 0.0026$ and given that $A_2 = 0.58$, $D_3 = 0$ and $D_4 = 2.115$. Compute UCL and LCL for arX and R chart. [7]

OR

What do you understand by ‘Statistical Quality Control’? Explain its need and importance in an industry. [14]